

HOMESTEAD MASTER PLAN

Optimized System for 2 People — Midlands, KwaZulu-Natal

Compressed edition: 11 sections, every fact retained

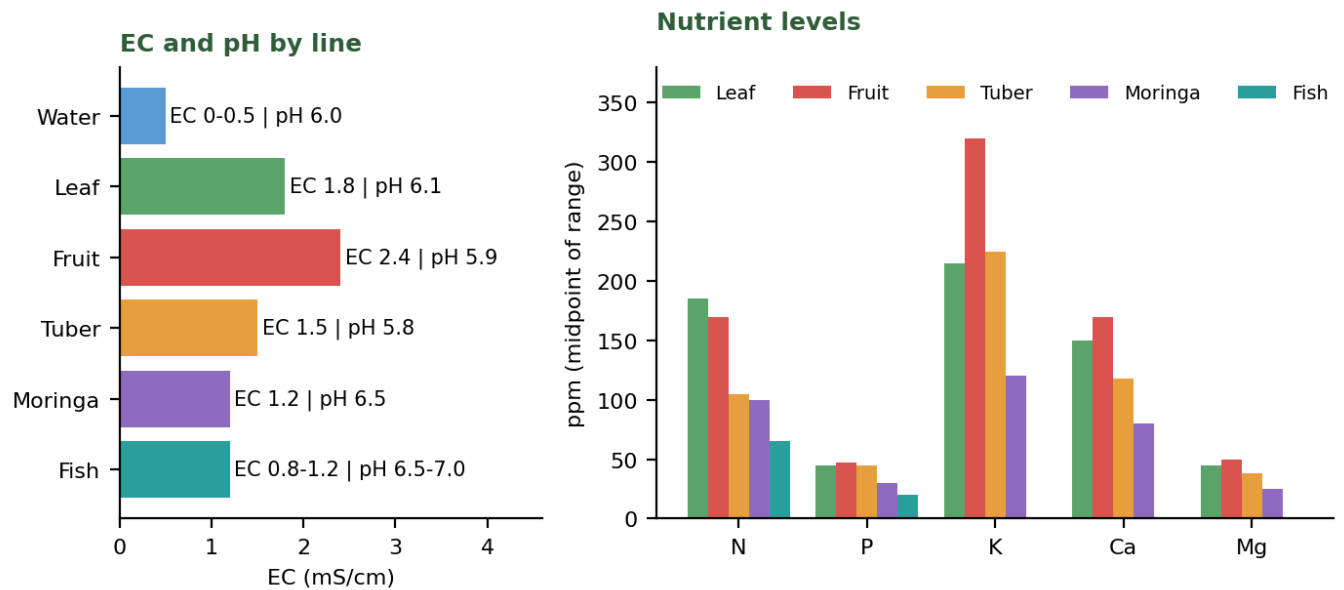
1. Sections & Loops

A section is a loop: one tank, one chemistry, all zones, mediums, animals, and humans it serves.

#	Loop	Chemistry	Zones	Mediums	Animals	Humans Get
1	Fish	Fish water (EC 0.8–1.2, pH 6.5–7.0)	Z1	Tanks, clarifier, biofilter, sump	Tilapia	—
2	Floating tray	Fish water	Z1	Floating trays	Duckweed, azolla (fed to tilapia)	—
3	Leaf	Leaf (EC 1.8, pH 6.1)	Z2, Z3, Z4	DWC raft + NFT gully + young Dutch buckets + isolated pot	Greens/herbs feed ducks, goats	Kale, chard, spinach, pak choi, coriander, basil, parsley, chives, spring onion, spearmint
4	Fruit	Fruit (EC 2.4, pH 5.9)	Z1, Z2	Fruiting Dutch buckets	Culls feed ducks, goats	Tomato, pepper, chilli, cucumber, zucchini
5	Tuber	Tuber (EC 1.5, pH 5.8)	Z1, Z4	Deep box + fabric bag	Vines feed ducks, goats	Sweet potato, potato
6	Moringa	Moringa (EC 1.2, pH 6.5)	Z1	40L pots	Prunings feed ducks, goats	Moringa leaf (dried, powdered)
7	Water	Water (EC 0–0.5, pH 6.0)	Z3	Nursery shelf	—	Seeds, cuttings
8	Duck	N/A	Z1	Deep litter; 30–40 m²	Ducks	Duck meat, duck eggs
9	Goat	N/A	Z4	Dry bedding; 20–30 m²	Goats	Goat meat, goat milk
10	BSF	N/A	Z1	Totes; 4–6 m²	BSF larvae	—
11	Mulberry	N/A	Outdoor	Soil; 400–600 m²	Mulberry	—

Space: GH ~230–330 m²; Outdoor ~1,300–1,850 m²; Total ~1,130–1,580 m² (duck/goat outdoor areas are the same mulberry hedges, counted once).

2. Chemistry



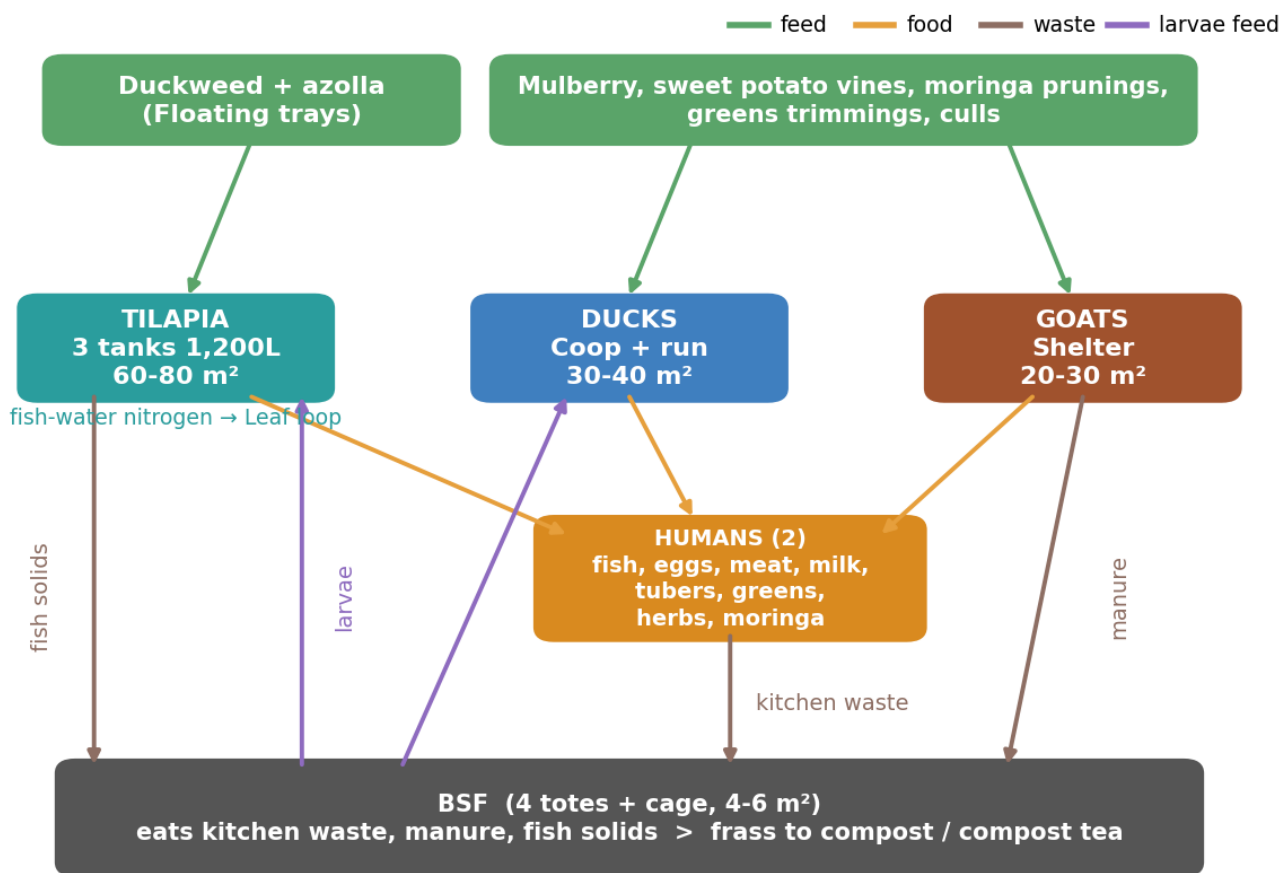
Line	N:K	Ca:Mg	Dosed?
Water	—	—	No
Leaf	0.85:1	3.2:1	Stock A+B + acid
Fruit	0.5:1	3.3:1	Stock A+B + acid + K
Tuber	0.45:1	3:1	Stock A+B + acid
Moringa	0.85:1	3.2:1	Stock A+B + pH up
Fish water	—	—	Bicarbonate only

Rule: Change pH by no more than 0.3 per day. Keep calcium apart from sulfate and phosphate (Stock A and B). Use Fe-DTPA or Fe-EDDHA.

Ranges (ppm): Leaf N 170–200, P 40–50, K 200–230, Ca 140–160, Mg 40–50. Fruit N 160–180, P 45–50, K 300–340, Ca 160–180, Mg 45–55. Tuber N 95–115, P 42–48, K 210–240, Ca 115–120, Mg 37–40. Moringa N ~100, P ~30, K ~120, Ca ~80, Mg ~25. Fish water N 30–100, P 10–30, K little. Water: all 0. Chart shows range midpoints.

Control system: EC/pH probe per loop; bleed valve, dilution valve, dosing pump, pH pump, K boost valve (Fruit), emergency isolation, backup aeration, manual override.

3. Animals



Animal	Housing / Space	Produces
Tilapia	3 tanks 1,200L; 60–80 m ²	Fish, nitrogen
Ducks	Coop + run; 30–40 m ²	Eggs, meat, manure
Goats	Shelter; 20–30 m ²	Milk, meat, manure
BSF	4 totes + cage; 4–6 m ²	Larvae, frass
Humans (2)	—	Kitchen waste, manure

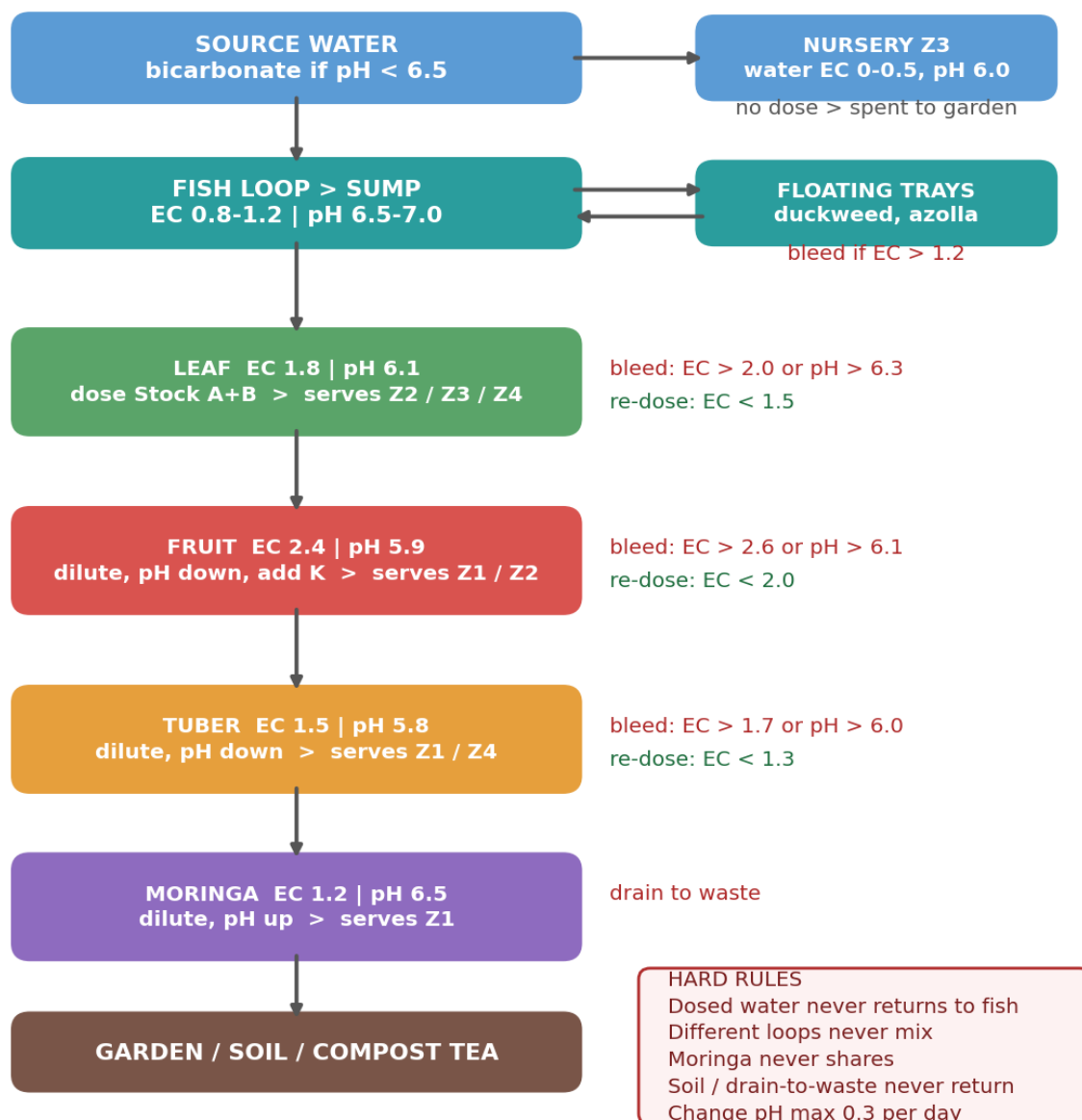
Humane slaughter: Tilapia — bolt pistol or percussive stun, then bleed. Ducks — non-penetrative captive bolt (convex head), then bleed. Goats — penetrative captive bolt, then bleed. Stun first, never bleed while conscious.

4. Feed Crops & Space

Species	Space	Zone	Medium	Loop	Feeds
Duckweed	40–60 m ²	Z1	Floating trays	Fish water	Tilapia
Azolla	10–15 m ²	Z1	Floating trays	Fish water	Tilapia
Mulberry	400–600 m ²	Outdoor	Soil, pollard hedges	N/A	Ducks, goats
Sweet potato vines (from tubers)	—	Z1+Z4	Coco:perlite 60:40	Tuber	Ducks, goats
Moringa prunings (from moringa)	—	Z1	Coco:perlite 60:40	Moringa	Ducks, goats

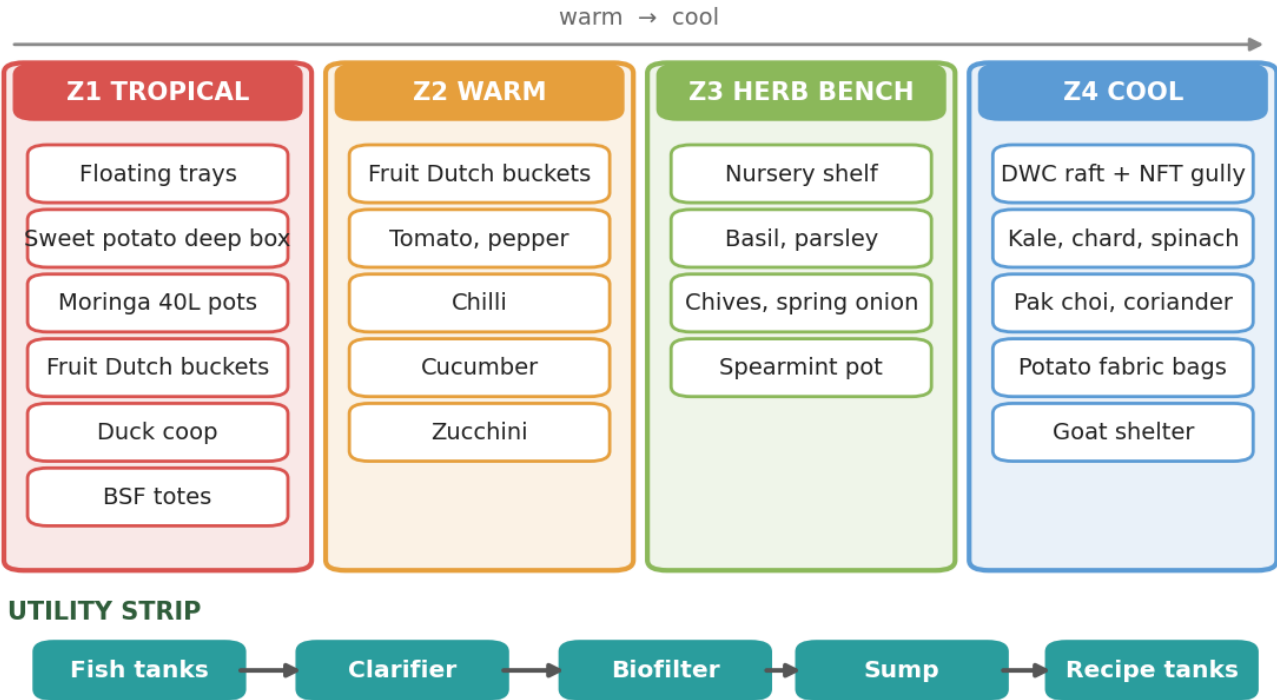
Mediums (8): Coco:perlite 60:40 (tubers, fruiting, moringa, spearmint); DWC raft + plug (greens); NFT gully + coco plug (herbs); floating trays (duckweed/azolla); water (tilapia); deep litter / dry bedding (ducks, goats); BSF totes; soil (mulberry).

5. Water Flow



Sequence: Dose at start, bleed when out of range, dilute if needed, re-dose only if EC drops. All spent water goes to garden / soil / compost tea. Soil never returns.

6. Layout



7. Plants: Stages & Staggered Sowing

Every	Species	Qty	Stage 1	Stage 2	Stage 3	Stage 4
1 wk	Pak choi	3	Water Z3	Leaf Z4	Leaf Z4	
1 wk	Spring onion	2	Water Z3	Leaf Z3	Leaf Z3	
2 wk	Spinach	6	Water Z3	Leaf Z4	Leaf Z4	Leaf Z4
2 wk	Coriander	2	Water Z3	Leaf Z4	Leaf Z4	
4 wk	Kale	3	Water Z3	Leaf Z4	Leaf Z4	Leaf Z4
4 wk	Swiss chard	3	Water Z3	Leaf Z4	Leaf Z4	Leaf Z4
4 wk	Sweet basil	2	Water Z3	Leaf Z3	Leaf Z3	
4 wk	Sweet potato	5	Water Z3	Tuber Z1	Tuber Z1	Tuber Z1 (flush)
4 wk	White potato	3	Water Z3	Tuber Z4	Tuber Z4	Tuber Z4 (flush)
5 wk	Zucchini	2	Water Z3	Leaf Z2	Leaf Z2	Fruit Z2
8 wk	Tomato	5	Water Z3	Leaf Z2	Leaf Z2	Fruit Z2
8 wk	Cucumber	2	Water Z3	Leaf Z2	Leaf Z2	Fruit Z2
12 wk	Bell pepper	6	Water Z3	Leaf Z2	Leaf Z2	Fruit Z2
13 wk	Parsley	3	Water Z3	Leaf Z3	Leaf Z3	Leaf Z3
13 wk	Moringa	2	Water Z3	Leaf Z1	Moringa Z1	Moringa Z1
16 wk	Chilli	2	Water Z3	Leaf Z2/Z1	Leaf Z2/Z1	Fruit Z2/Z1
26 wk	Garlic chives	2	Water Z3	Leaf Z3	Leaf Z3	Leaf Z3
26 wk	Spearmint	2 pots	Water Z3	Leaf Z3	Leaf Z3	Leaf Z3
Continuous	Duckweed	half / 3 days	Fish Z1	Fish Z1	Fish Z1	Fish Z1
Continuous	Azolla	half / 3 days	Fish Z1	Fish Z1	Fish Z1	Fish Z1

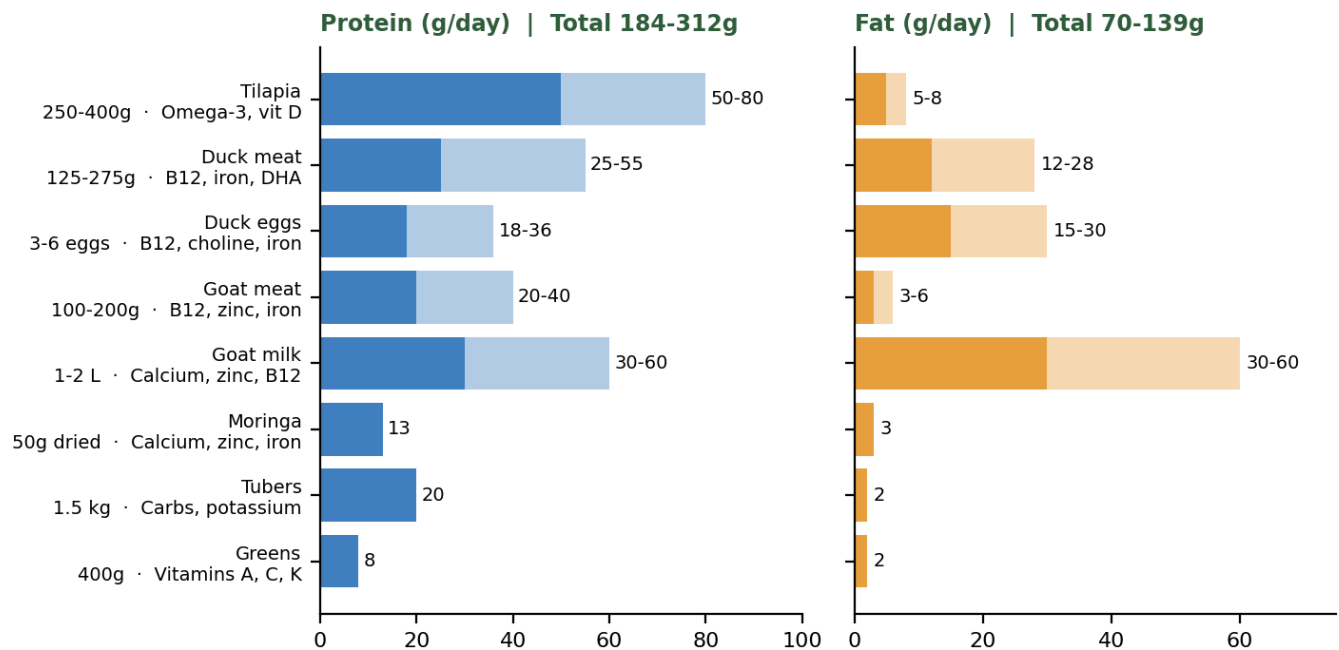
Water
Leaf
Fruit
Tuber
Moringa
Fish

Seed quantities per sowing: Pak choi 5–6; spring onion 4–5; spinach 10–12; coriander 4–5; kale 5–6; chard 5–6; basil 4–5; tomato 8–10; cucumber 4–5; pepper 10–12; chilli 4–5; zucchini 4–5; parsley 5–6; moringa 4–5; chives 4–5; spearmint cuttings.

Animal rhythm: Every 1 wk — new BSF tote, collect eggs daily. Every 2 wk — harvest half duckweed/azolla. Every 6 wk — breed next goat doe. Every 8 wk — add 35 tilapia, harvest from tank 3. Every 26 wk — replace 2 ducks. Continuous — ducks breed, goats kid yearly.

Week rhythm: Mon sow/plug; Tue move batches to next loop; Wed EC/pH check every tank; Thu harvest; Sat cut animal greens and prune moringa.

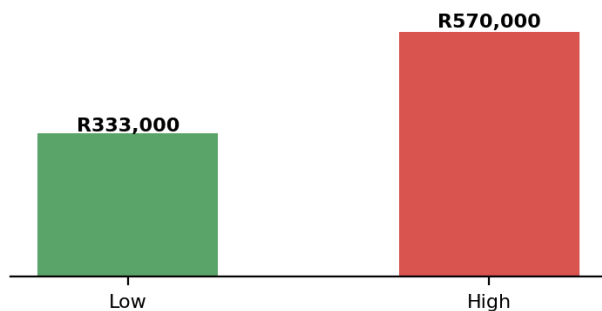
8. Diet (2 People, Per Day)



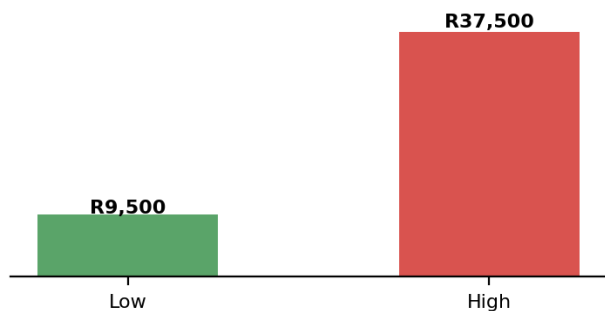
Targets met: 2,500 kcal, 150g protein, ~312g carbs, 10%+ fat. Moringa covers calcium and zinc. Animal foods cover B12, iron, omega-3.

9. Cost & Solar

Capital (DIY, solar, greenhouse)



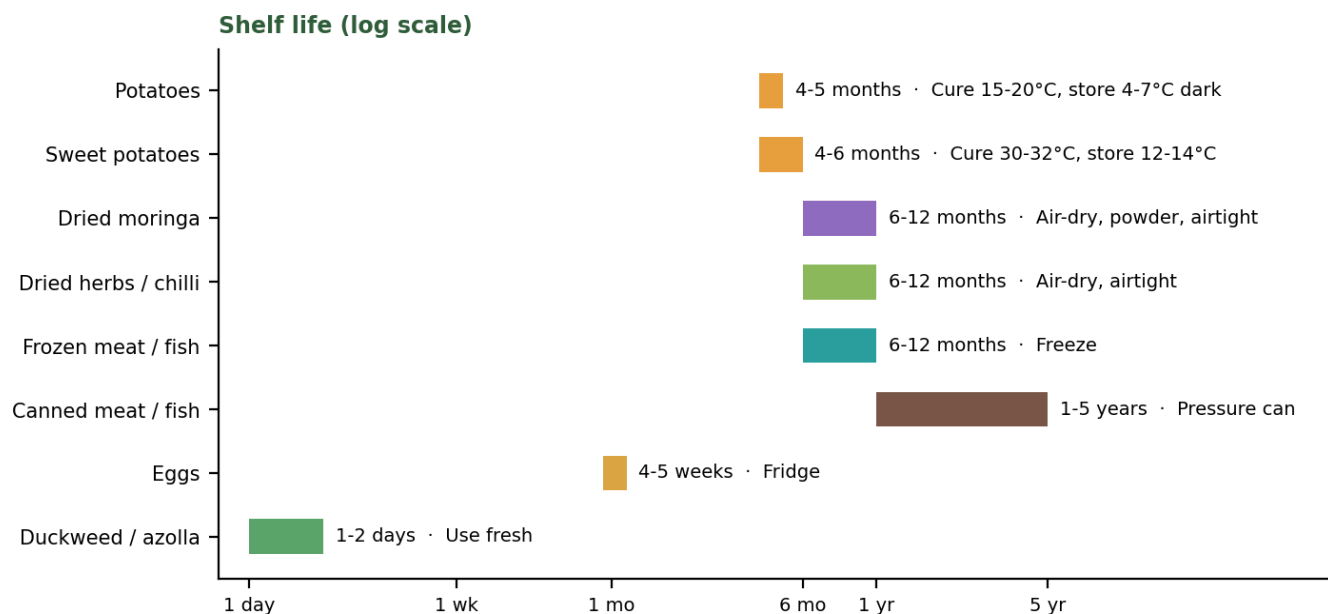
Annual running



Solar load: Winter 12–21 kWh/day; Summer 2–4 kWh/day. Recommended: 8–10 kWp panels + 20 kWh battery + 5kW inverter + CoC. Generator backup 3kW.

Buy list: Salt + iodine (cannot grow); cooking oil (unless oilseed crops added); vitamin D3 winter (sunlight insufficient); spices (black pepper, cumin).

10. Storage



11. Summary

Structure: 11 sections = 7 loops + 4 animal/fodder sections. 6 chemistry lines, 8 mediums, 4 zones + outdoor, 15+ plants, 4 animals, 2 humans fed. Total ~1,130–1,580 m² (GH ~230–330 m²).

Water: Fish > Floating trays > Fish. Fish > Leaf > Fruit > Tuber > Moringa > Soil. Dose at start, bleed when out of range, dilute if needed, re-dose only if EC drops. Nothing dosed returns to fish. Soil never returns.

Animals: Tilapia in fish loop. Ducks and goats eat trimmings, produce eggs/milk/meat/manure. BSF converts waste to feed. Humans eat everything; waste feeds BSF.

Output: Feeds 2 people 2,500 kcal and 150g protein daily. All animal feed grown on site. Only salt, iodine, oil, and winter vitamin D bought.